Tokenized Time-Series in Satellite Image Segmentation With Transformer Network for Active Fire Detection

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***Abstract*— The Visible Infrared Imaging Radiometer Suite (VIIRS) onboard the Suomi National Polar-orbiting Partnership (Suomi-NPP) satellite has been used for the early detection and daily monitoring of active wildfires. How to effectively segment the active fire (AF) pixels from VIIRS image time-series in a reliable manner remains a challenge because of the low precision associated with high recall using automatic methods. For AF detection, multicriteria thresholding is often applied to both low-resolution and mid-resolution Earth observation images. Deep learning approaches based on convolutional neural networks (ConvNets) are also well-studied on mid-resolution images. However, ConvNet-based approaches have poor performance on low-resolution images because of the coarse spatial features. On the other hand, the high temporal resolution of VIIRS images highlights the potential of using sequential models for AF detection. Transformer networks, a recent deep learning architecture based on self-attention, offer hope as they have shown strong performance on image segmentation and sequential modeling tasks within computer vision. In this research, we propose a transformer-based solution to segment AF pixels from the VIIRS time-series. The solution feeds a time-series of tokenized pixels into a transformer network to identify AF pixels at each timestamp and achieves a significantly higher F1-score than prior approaches for AFs within the study areas in California, New Mexico, and Oregon in the U.S., and in British Columbia and Alberta in Canada, as well as in Australia, and Sweden.**

***Index Terms*— Active fire (AF) detection, image segmentation, remote sensing, transformer, Visible Infrared Imaging Radiometer Suite (VIIRS).**

I. Introduction

In RECENT years, wildfires have been increasing in frequency and severity due to the hotter and drier conditions in many areas caused by human-induced climate change [1], [2]. Extensive studies in active fire (AF) detection have been made with different satellite platforms and algorithms [3], [4], [5], [6]. Based on existing AF products, NASA published the wildfire monitoring platform Fire Information for Resource Management System (FIRMS) [7]. AF products using Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS) are mainly involved within the platform. MODIS AF product (MCD14DL) and VIIRS AF product (VNP14IMG) both offer twice daily revisit but VIIRS AF product has 375-m spatial resolution and MODIS AF product has 1-km spatial resolution [8]. In [9], the survey introduces the history of MODIS and VIIRS AF products. MODIS is the sensor belonging to NASA's Earth Observing System [10]. It contains two mid-infrared channels with different saturation temperatures to help detect AF. VIIRS is the sensor onboard the National Polar-orbiting Partnership (Suomi-NPP) satellite, it also provides two mid-infrared channels, but they have different spatial resolutions 375 and 750 m. The AF products of both sensors utilize multicriteria threshold-based methods to detect AF pixels. The MODIS AF product is introduced in [10]. Giglio et al. [11] improved the contextual algorithms by using larger window sizes to dynamically change the thresholds. Currently, MODIS Collection 6 AF detection algorithm is still operating [6]. Similarly, AF product based on VIIRS is also developed [12]. AF products based on MODIS and VIIRS are utilized in mapping burned areas by aggregating the AF points detected during each timestamp [3], [13]. While these products are very useful, the existing solutions have flaws, including many false alarms due to cloud cover or buildings with high-temperature roofs. Also, the multicriteria threshold-based method does not use rich temporal information of each pixel at different timestamps and rich spatial information between neighboring pixels. To improve the accuracy of AF detection, deep learning as a powerful tool for spatial and temporal modeling is introduced in AF detection. Deep learning is widely used in analyzing remote sensing big data [14] and applied in wildfire monitoring. Spatial deep learning models like convolutional neural networks (ConvNets) are used to model the spectral and spatial characteristics of the remote sensing images to detect the burned area or AF pixels. For AF detection, de Almeida Pereira et al. [15] published an AF dataset and showed that ConvNets can be trained to detect AFs. Burned area mapping is also well studied [16], [17], [18] with ConvNets. There is limited research, however, on how to best exploit the temporal information available from the sequence of remote sensing images from a location to detect the AF more accurately. In [19], it was shown that recurrent neural networks (RNNs) can be used to detect AFs from optical multispectral Earth observation images with coarse spatial

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resolution. Using higher spatial resolution with lower temporal resolution images is possible, but complicates the processing required as it needs to deal with the variations of the multiple spectral bands during the wildfire and the longer time gap between captured images. To better model the fluctuation of the spectral bands, transformer network is used in this article. Transformer network is a state-of-the-art sequential deep learning model, widely applied in time-series analysis and natural language processing (NLP). With the self-attention mechanism, transformer network can generate a more robust representation than RNN and provide better classification performance.

The objective of this research is to improve AF detection accuracy using a time-series deep learning model from a multithreshold-based AF product. In this article, we assume to have access to a sequence of spatially aligned satellite images with multiple spectral bands. We introduce a simple but effective transformer network approach to predict fire/nonfire from this data. The network analyzes the time-series generated by the spectral band data to make its predictions in the following manner. A pixel location at each time step is represented by a vector extracted from its spectral bands' values plus an encoding of its temporal position within the sequence. The feature vector sequence, from one spatial location, is fed into the transformer network, which then predicts for each time step fire or nonfire. The process is then repeated independently for each spatial pixel location with the same trained network. The network is trained on a dataset of 19 large wildfires, which occurred in 2020 from the Suomi-NPP VIIRS satellite images with labels manually created. The test set consists of 18 large wildfires occurring in California, Oregon, and New Mexico in the U.S., British Columbia and Alberta in Canada, New South Wales in Australia, and Sweden not included in the training set. The ground truth annotations for the test set were acquired through manual inspection.

We report the results of comprehensive ablation experiments investigating the transformer architecture (depth and width) and the parameters controlling the feature representation of an individual pixel. In addition, we also compare the results obtained by our transformer model to other standard deep learning architectures—those relying solely on spatial information with ConvNet architectures and others relying on sequential data only with long-short term memory (LSTM) and gated recurrent unit (GRU) architectures. Our results highlight the importance of temporal over spatial information for AF detection when using VIIRS input as the networks based on temporal data completely dominate and outperform by a large margin those relying solely on spatial information. For the sequential-based networks, the performance differences between architectures are smaller, but the transformer consistently has a higher level of performance. We also compare the result to the baseline of the VIIRS AF product provided by NASA FIRMS and show that the temporal-based deep learning models with our design dominate.

In summary, there are three main contributions.

1) We present a transformer-based method that can more accurately detect AF compared to the VIIRS AF products. The proposed transformer-based method uses the rich temporal information of the VIIRS time-series and can also outperform ConvNet-based models and other classic temporal models like LSTM and GRU.
2) We present a novel way to treat time-series of low spatial resolution Earth observation images, more specifically, tokenize images by utilizing the temporal information. Tokenization methods enable temporal models to have fewer number of parameters than ConvNet models and substantially outperform other spatial models for AF detection.
3) A wildfire dataset generated from Suomi-NPP VIIRS 153 images consisting of five bands at 375-m resolution as shown in Fig. 1.

II. Related Work

A. Semantic Segmentation for EO Images

Deep learning models for semantic segmentation are widely utilized in Earth observation applications. The major objective of utilizing semantic segmentation models on Earth observation data is to extract information or detect targets or anomalies. [20], [21], [22], [23]. In recent years, semantic segmentation has been dominated by ConvNets. The fully convolutional network (FCN) [24] is an early representative work of semantic segmentation. The architecture of the FCN, as the name indicates, contains only convolutional layers and achieves pixelwise prediction by upsampling the encoded representation to the input size. With a similar encoder–decoder architecture, U-Net [25] concatenates features at different scale levels with a skip connection between downsampling and upsampling layers and thus enables its output segmentation map to preserve the high-level texture of the input images. The U-Net's simple architecture frequently allows it to outperform other networks, especially on the small dataset—a common scenario for many Earth observation and medical image applications. The success of U-Net has promoted many U-Net variants [26], [27], [28]. One such example is LinkNet [29] which combines features between encoding and decoding layers via addition as opposed to concatenation and replaces convolution blocks both in the encoder and decoder with ResNet blocks [30]. PSPNet [31] proposes a pyramid pooling module to embed context information at different levels to help the final segmentation. A network architecture based on a similar idea is the feature pyramid network (FPN) [32] which efficiently extracts multiscale feature representation of the input image in a process independent of the backbone ConvNet architecture.

As for wildfire applications, there are many systems based on ConvNet-based semantic segmentation networks. In [18], a ConvNet network is proposed to segment burned areas using Sentinel-1 images. Similarly, utilizing optical images of Sentinel-2, Knopp et al. [16] also proposed a U-Net method to segment burned area. For AF detection, Pereira et al. [33] published a dataset based on Landsat-8 images and evaluated the performance of U-Net on the dataset. Noticeably, this article shares similar objectives with [33], but the main source of data behind is different, both in terms of spatial resolution

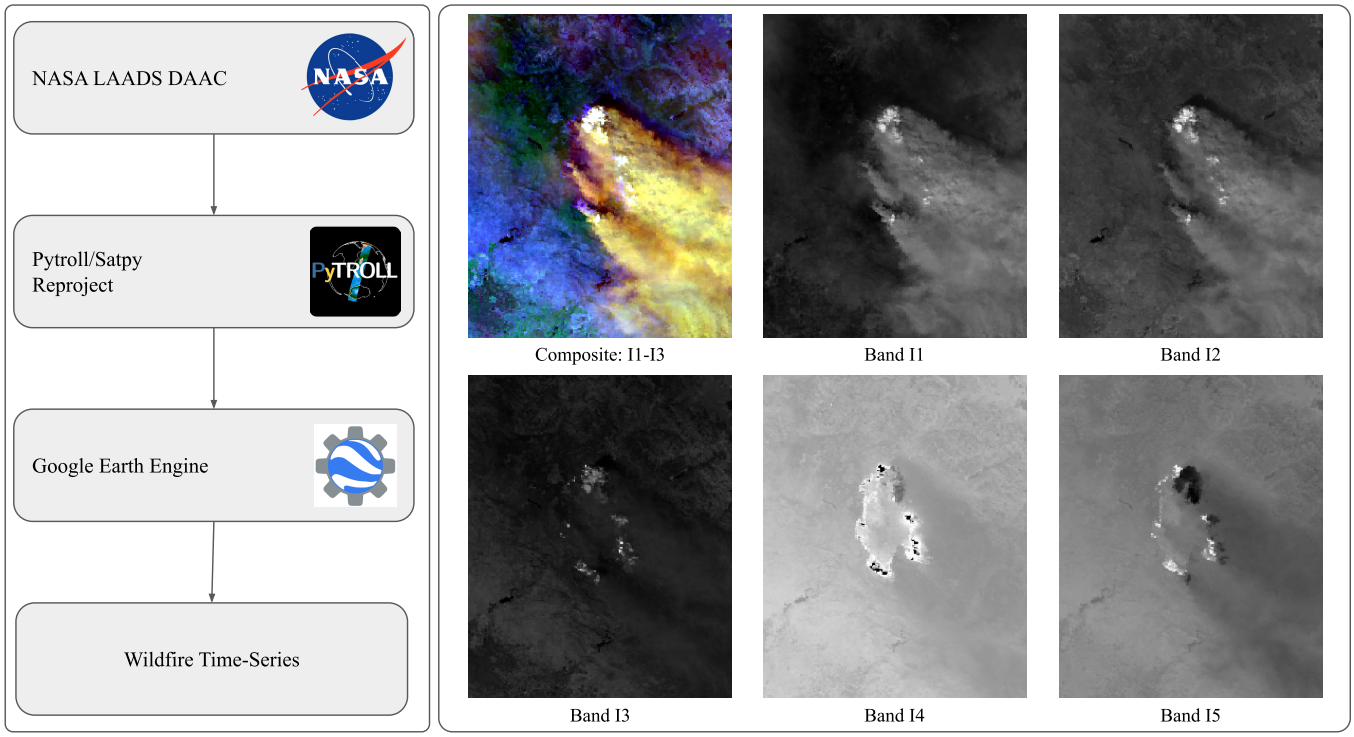


Fig. 1. (a) Pipeline applied to acquire the aligned data used by the networks in this article. (b) Visualization of the five different electromagnetic bands available at each time step. The data shown is taken from the Creek Fire, CA, USA, September 7, 2020. The composite image is the false color composite of Band I1, Band I2, and Band I3.

and temporal resolution. The satellite platform of VIIRS offers one day image every day while Landsat has a revisit time of 16 days. VIIRS only has a 375-m spatial resolution compared to Landsat OLI. We argue that frequent revisit is more essential in monitoring AFs than spatial resolution in practice, as agile detection can help decision makers to respond quickly.

B. Transformer

Transformer models recently have shown strong improvement in NLP tasks [34], [35], [36] and increasingly extend their impact on Computer Vision tasks [34], [37]. Owing to its self-attention mechanism, the representation generated by the encoder is contributed by every pixel in the images, which generates a stronger representation than ConvNet-based methods. In recent works, transformer also actively is involved in semantic segmentation tasks [38], [39], [40], [41]. Based on the classic U-Net architecture, Chen et al. [40] used the transformer to generate a stronger representation at latent space. In [41], the proposed architecture changes all the ConvNet layers to swin-transformer layers and provides a transformer-based segmentation model which shows promising results for medical images. The encouraging results indicate representation generated using self-attention empowers the network to produce better segmentation results. The transformer network provides a significant advantage in dealing with time-series data. To deal with time-series of images, Arnab et al. [42] proposed a transformer model which consists of multiple spatial encoders and temporal encoders to deal with time-series data.

For remote sensing applications, the transformer network is also a popular architecture. Garnot et al. [43] proposed a transformer network to encode a set of pixels to classify crop types using time-series of Sentinel-2 images. In [44], a transformer-based method is applied for building change detection purposes. Similarly, Bazi et al. [45] applied the vision transformer proposed in [37] to classify remote sensing images. As for remote sensing image segmentation tasks, there are several works on applying different transformer networks to extract context information from Earth observation images. Xu et al. [46] proposed a model based on the swin transformer, which shows optimal results in the segmentation results. Moreover, on the wildfire side, Ghali et al. [47] also applied a vision transformer to segment burning wildfires, which shows great potential to detect AFs. However, we argue that the data source used in [47] are not Earth observation images, and is also not multispectral.

There is limited work on wildfire applications using Earth observation images and transformer networks. We argue [43] shows similarity to our work in the methodology part, because both proposed methods are applied to time-series Earth observation images. Both methods also have individual pixels as input. But the two models serve two different purposes, the method proposed in this manuscript is intended to segment the image, while the network of [43] is trained for image classification. Also, the tokenization strategy in this work is different from the method proposed by Garnot et al. [43]. While we try to classify all the pixels in the images to create a segmentation map of AF pixels, Garnot et al. [43] took randomly sampled pixels to classify the whole image.

III. Methodology

The transformer model architecture is shown in Fig. 2, it closely follows the design of the original encoder of the original transformer [35] and the vision transformer [37]. The major difference to the ViT paper is that our input to the transformer is a sequence of vectors representing one spatial location over time, as opposed to a sequence of image patches representing the different positions in one image. The pipeline for AF classification has three high-level steps.

1) Tokenize the sequence of images into WH time-series, where each time-series is a sequence of tokens representing one spatial location over time and each token is combined with an embedding of its temporal position in the time-series.
2) Encode each time-series with a masked transformer encoder and classify the pixel location at each time stamp with a layer of multilayer perceptron (MLP) as AF or not.
3) Obtain a final sequence of predictions for the original image sequence of AF or not.

A. Tokenization of VIIRS Image Sequence

Transformer networks, in contrast to convolutional networks, facilitate efficient modeling of long-range dependencies. For the problem tackled in this article, we have a temporal sequence of low-resolution VIIRS images. Each pixel in a VIIRS image corresponds to 375 m on the ground. Thus the spatial correlation between neighboring pixels in a VIIRS image will, in general, be much lower than for regular images as the ground land cover can change significantly in a 375 × 375 m area. The frequent revisit of the Suomi-NPP, fortunately, results in relatively high temporal resolution imaging of the area and this compensates for the low spatial resolution of VIIRS. With this type of data much of the information about the AF is given by temporal changes at each spatial location as opposed to variations across the spatial dimensions (confirmed by the experimental results in Section V). To exploit the temporal information to detect where the AF is present, we perform the following. Each VIIRS image X has size $W \times H$ with C channels and we have $X_1, \ldots, X_\tau$ VIIRS images in a given sequence. For each pixel location (x, y) we extract an image patch of size $w \times w$ centered at (x, y) from each X_t with $1 \leq t \leq \tau$. We denote the extracted image patches as X_{xyt}. We then tokenize the image patch, X_{xyt} by flattening it to obtain a vector x_{xyt} of size $w^2C \times 1$. It is this sequence $x_{xy1}, x_{xy2}, \ldots, x_{xy\tau}$ which is the input to our transformer network. Note we generate such a sequence for every pixel (x, y). For the rest of this article when we refer to the input of sequence vectors we drop the xy subscripts to reduce the notation clutter. In the experiments, different values of w (corresponding to different-sized image patches) are investigated and its impact on the performance is discussed in Section V-B3.

As shown in Fig. 3, an AF progresses locally, and the value of Band I4 significantly increased when an AF happened. In contrast, for regions without AF, the value remains stable.

B. Model Architecture

As shown in Fig. 2, the encoder network we use has the same high-level architecture as the transformer [35] encoder, except that the class token is not included. The encoder consists of multiple stacked transformer layers. Each transformer layer has the same hyperparameters such as the number of heads of the multihead attention layer and hidden units in the MLP layer.

Inside each transformer layer, there are four internal layers plus skip connections between layers. For completeness and transparency we review the mathematics of the standard transformer layer [35], [37]. Denote one input token into a transformer layer as $f_t^{(l)} \in \mathbb{R}^k$ from one training example. Then one can gather all the input tokens to the layer as

$$F_l = \begin{bmatrix} \leftarrow f_1^{(l)T} \rightarrow \\ \vdots \\ \leftarrow f_\tau^{(l)T} \rightarrow \end{bmatrix} \in \mathbb{R}^{\tau \times k}. \tag{1}$$

As shown in Fig. 2 (right), layer normalization is applied to the output from the previous layer

$$\tilde{f}_t^{(l)} = \mathrm{LN}\left(f_t^{(l)}\right). \tag{2}$$

Layer Normalization [48] at layer l calculates the mean $\mu_t^{(l)}$ and standard deviation $\sigma_t^{(l)}$ for $f_t^{(l)}$

$$\mu_t^{(l)} = \frac{1}{k}\sum_{i=1}^{k} f_{ti}^{(l)}, \quad \sigma_t^{(l)} = \sqrt{\frac{1}{k}\sum_{i=1}^{k}\left(f_{ti}^{(l)} - \mu_t^{(l)}\right)^2}. \tag{3}$$

The normalized responses are then scaled and shifted by the learned parameters $\gamma_t^{(l)} \in \mathbb{R}^k$ and $\beta_t^{(l)} \in \mathbb{R}^k$

$$\tilde{f}_t^{(l)} = \gamma_t^{(l)} \odot \left(\frac{f_t^{(l)} - \mu_t^{(l)}}{\sigma_t^{(l)}}\right) + \beta_t^{(l)} \tag{4}$$

where $\odot$ represents elementwise multiplication. Layer Normalization results in more stable training reduce training times and provide better generalization abilities for transformer and RNNs.

After layer normalization, multihead self-attention is applied. Assume there are A self-attention heads. For one attention head $a \in \{1, \ldots, A\}$ the layer normalized features are multiplied separately by three parameter matrices $W_{kla}, W_{qla}, W_{vla} \in \mathbb{R}^{k \times k/A}$ to calculate keys, queries, and values $K_{la}, Q_{la}, V_{la} \in \mathbb{R}^{\tau \times k/A}$ which are in turn used to calculate self-attention in the following manner. Gather the normalized features $\tilde{f}_t^{(l)}$ in $\tilde{F}_l$ similar to (1). The key, query, and value matrices for one head are computed as

$$\begin{aligned} K_{la} &= \tilde{F}_l W_{kla} \\ Q_{la} &= \tilde{F}_l W_{qla} \\ V_{la} &= \tilde{F}_l W_{vla}. \end{aligned} \tag{5}$$

The similarity between a query vector q, a row of Q_{la}, and a key vector corresponding to a row of K_{la}, is computed as a dot product. The attention between q and each key vector is computed as $\mathrm{Softmax}(q K_{la}^T)$. To avoid the increasing peakiness of the output of softmax as the dimension of the key and query vectors grows, Vaswani et al. [35] suggested

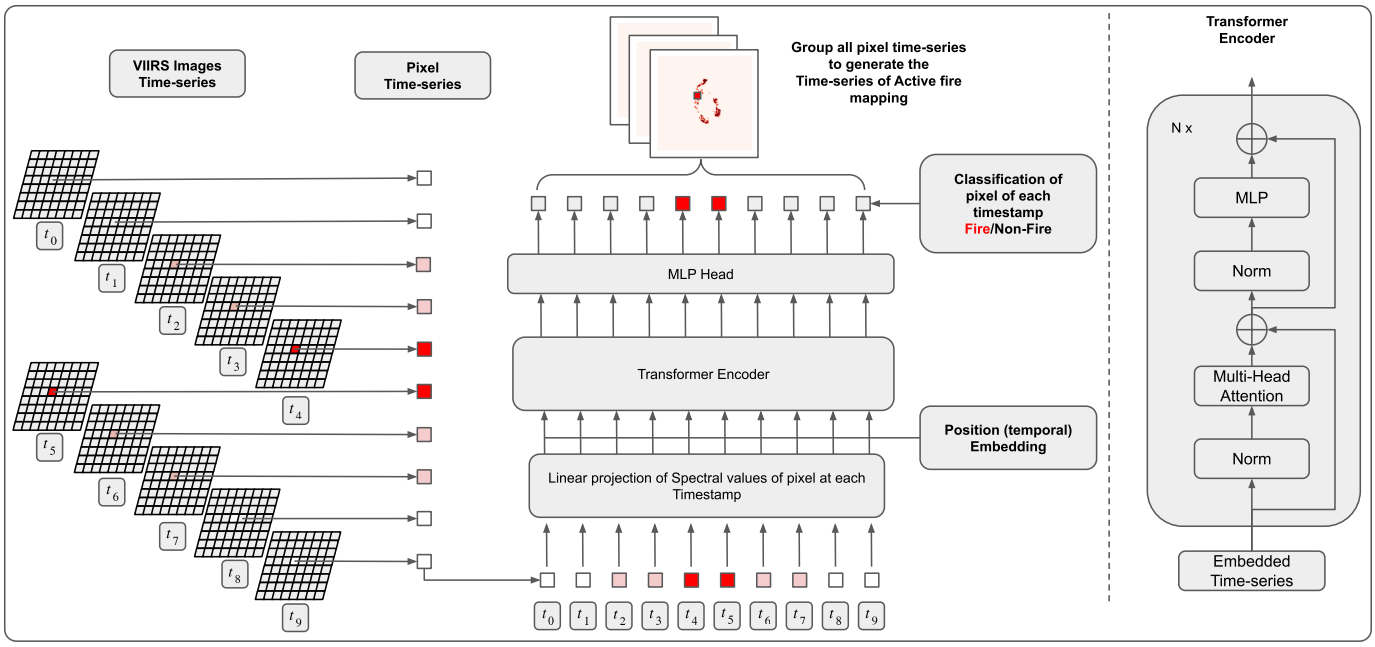


Fig. 2. Tokenized time-series in satellite image segmentation with transformer network for AF detection.

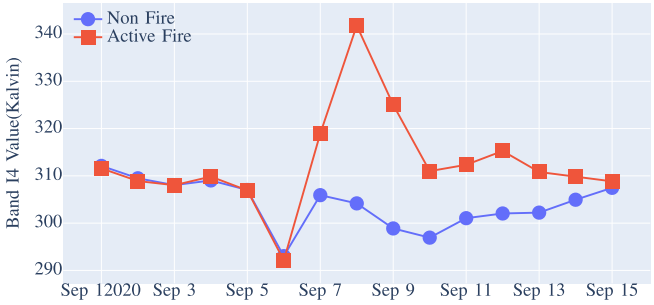


Fig. 3. Fluctuation of values of Band I4 for two pixel locations. One is a nonfire pixel (blue) and the other (red) is an AF location on September 8, 2020.

scaling the dot-product by $(k/A)^{1/2}$. Thus the attention between every pair of queries and key vector stored in Q_{la} and K_{la} is computed as

$$\Lambda_{la} = \text{Softmax}\left(\frac{Q_{la}K_{la}^T}{\sqrt{k/A}}\right) \in \mathbb{R}^{\tau\times\tau} \tag{6}$$

where Softmax is applied independently to each row. Then the output is computed as

$$\text{Attention}(Q_{la}, K_{la}, V_{la}) = \Lambda_{la}V_{la} = F'_{la} \in \mathbb{R}^{\tau\times k/A}. \tag{7}$$

For multihead attention layers of the classic transformer encoder, the multihead attention layer is applied to all input tokens. This means all input tokens contribute to the calculation of the attention score of one token. But our AF detection network has a time-series of pixels as input, and the model should only use past and current tokens for classification at the current frame t. Thus, in the attention module, masked self-attention is applied. As shown in (8), for self-attention with masking, the attention between every pair of queries Q_{la} and key vectors K_{la} is masked by a matrix M_{la} of size $\tau \times \tau$. All entries of M_{la} on and below the main diagonal have value 1 and value $-\infty$ for all the entries above the main diagonal. After masking all the entries of $Q_{la}K_{la}^T$ above the main diagonal are set to $-\infty$. Then

$$\Lambda_{la}^{\text{masked}} = \text{softmax}\left(\frac{Q_{la}K_{la}^T}{\sqrt{k/A}} \odot M_{la}\right) \in \mathbb{R}^{\tau\times\tau} \tag{8}$$

where $\odot$ represents elementwise multiplication. This process ensures all the entries in Λ_{la} above the main diagonal are zero and in turn ensures only representations from the current and previous timestamps are used to generate the representation for time t at layer l. Thus, no information leaks from future tokens. After each of the multihead self-attention heads has been applied, the outputs from all the heads for each time step are concatenated and are linearly transformed

$$f'^{(l)}_t = \text{Linear}\left(\begin{bmatrix} f'^{(l)}_{t1} \\ \vdots \\ f'^{(l)}_{tA} \end{bmatrix}\right) + f^{(l)}_t. \tag{9}$$

Following the multihead self-attention, the final output of the transformer layer is generated after another layer normalization and a two-layer MLP, with M denoting the dimensionality of the hidden layer of this MLP, is applied plus a skip connection:

$$f^{(l+1)}_t = \text{MLP}\left(\text{LN}\left(f'^{(l)}_t\right)\right) + f'^{(l)}_t. \tag{10}$$

Finally a prediction, of the pixel being an AF pixel, for each time-step $1 \le t \le \tau$ is produced via a simple linear layer and applying the sigmoid function

$$y_t = \sigma\left(\text{Linear}\left(f^{(L)}_t\right)\right) \in [0, 1]. \tag{11}$$

IV. Dataset

A. Satellite Sensor Description

AF detection requires electromagnetic bands that are sensitive to high ground temperatures according to Wien's

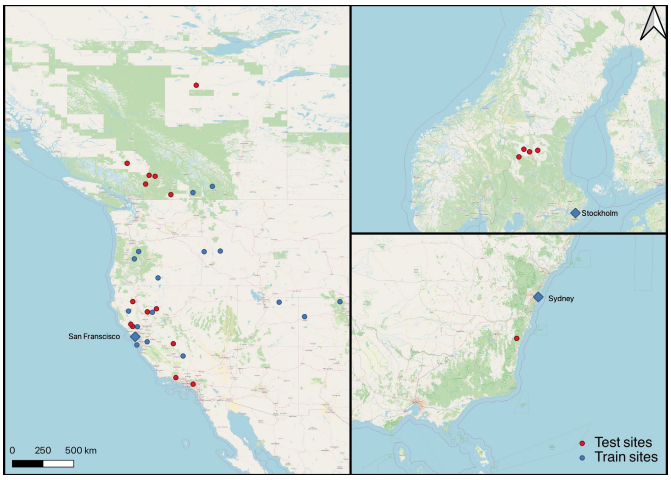


Fig. 4. Geolocation of the wildfires in the training and test sets.

TABLE I
VIIRS IMAGERY BANDS AND SPECTRUM

Bands	Wavelength range (μm)	Description
Band I1	0.60 – 0.68	Red
Band I2	0.85 – 0.88	Near Infrared
Band I3	1.58 – 1.64	Short-wave Infrared
Band I4	3.55 – 3.93	Medium-wave Infrared
Band I5	10.50 – 12.40	Long-wave Infrared

Law. The burning biomass often generates high spectral radiance which has a wavelength located in the middle infrared (MIR) [49]. The VIIRS, carried by the Suomi-NPP Satellite, enables detecting AF with the MIR band as it provides twice daily revisits and 375-m spatial resolution. Both of the alternative satellites, MODIS and Sentinel-3 Sea and Land Surface Temperature Radiometer (SLSTR) have the same temporal resolution but lower spatial resolution. The higher spatial resolution of VIIRS images makes them more suitable to detect and locate AFs more accurately than the other available alternatives.

B. Dataset Collection

The images are collected from the VIIRS radiance (VNP02IMG) and geolocation (VNP03IMG) products. They are then processed by the SatPy library [50] and Google Earth Engine to generate time-series of wildfire images [51]. The processing pipeline is presented in Fig. 1(a). The images have five bands with 375-m spatial resolution. Details of the wavelength bands are provided in Table I. According to Wien's Law, the blackbody radiation curve peak at different wavelengths based on the temperature. From Fig. 1(b), we see that the short-wave infrared Band I3, medium infrared Band I4 and long-wave infrared Band I5 can all contribute to detecting AFs, but the majority of AF pixels are the most distinctive in the medium infrared Band I4. Another important factor to consider is the interference of clouds and smoke which makes AF detection challenging. When either smoke or cloud is thick, AF pixels can be misclassified as clouds. Cloud and smoke are detectable in all bands, but are more distinctive in the red Band I1 and Infrared band I2.

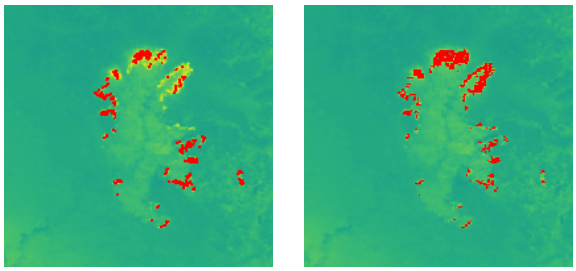
TABLE II
WILDFIRE TRAINING SITES

Fire Name	Coordinates	Start Date	End Date
Training Sites			
California, US			
August Complex Fire	(-122.97, 39.87)	2020-08-15	2020-09-16
LNU Lightning Complex Fire	(-122.24, 38.59)	2020-08-16	2020-10-02
SCU Lightning Complex Fire	(-121.44, 37.35)	2020-08-16	2020-10-01
CZU Lightning Complex Fire	(-122.28, 37.10)	2020-08-16	2020-09-22
North Complex Fire	(-121.01, 39.78)	2020-08-14	2020-09-13
Glass Fire	(-122.50, 38.57)	2020-09-27	2020-10-30
Castle Fire	(-118.50, 36.19)	2020-08-20	2020-09-15
Oregon, US			
Beachie Creek Fire	(-122.14, 44.75)	2020-08-17	2020-12-10
Holiday Farm Fire	(-122.49, 44.15)	2020-09-06	2020-10-29
Brattain Fire	(-120.55, 42.59)	2020-09-05	2020-09-15
Washington, US			
Cold Spring Fire	(-119.57, 48.85)	2020-09-06	2020-09-15
Colorado, US			
Cameron Peak Fire	(-105.65, 40.64)	2020-08-13	2020-08-30
Pine Gulch Fire	(-108.55, 39.42)	2020-08-08	2020-08-18
Montana, US			
Drumming Fire	(-113.16, 48.15)	2020-08-26	2020-09-06
Utah, US			
East Fork Fire	(-110.63, 40.60)	2020-09-02	2020-09-15
Idaho, US			
Woodhead Fire	(-116.77, 44.74)	2020-09-05	2020-09-15
Buck Fire	(-115.45, 44.79)	2020-08-26	2020-09-06
BC, Canada			
Doctor Creek Fire	(-116.10, 50.09)	2020-08-18	2020-09-21
Talbott Creek	(-117.69, 49.59)	2020-08-22	2020-09-02
Validation Sites			
BC, Canada			
Hanceville Fire	(-123.00, 51.98)	2017-07-07	2017-07-30
California, US			
Walker Fire	(-120.68, 40.06)	2019-09-04	2019-10-24

The new dataset collected for this study is from the fire season of 2020 as shown in Table II. It consists of the day images from 19 large wildfires between August 2020 and September 2020 in North America. As night images only have the spectral bands I4 and I5 we omit them from the dataset. As shown in Fig. 4, all the images are collected from NASA LAADS DAAC (Level-1 and Atmosphere Archive and Distribution System Distributed Active Archive Center), reprojected locally with the SatPy Python package, processed and cropped in Google Earth Engine. After processing, images at each time stamp have size 224×224 with the AF roughly located in the center of the image. The temporal duration of each wildfire varies, but each wildfire sequence is partitioned into subsequences of length 10 which represents VIIRS images of ten consecutive days. As a consequence, the training dataset consists of 58 time-series of AF image sequences of length 10. Two wildfire sites from two different years are used as the validation dataset to complete model selection. The testing dataset consists of 18 wildfires across multiple continents.

C. Dataset Annotation

One challenge of this application is getting accurately labeled training data on active wildfires. The VIIRS 375-m



(a) (b)

Fig. 5. Qualitative example of the higher quality of our manual labels compared to those produced automatically by VIIRS AF product. The red pixels overlaid on each Band I4 image denote the pixels labeled as AF pixels. Visualization of the AF labels produced by (a) VIIRS AF product and (b) our manual inspection. In the former many of the AF pixels are missing.

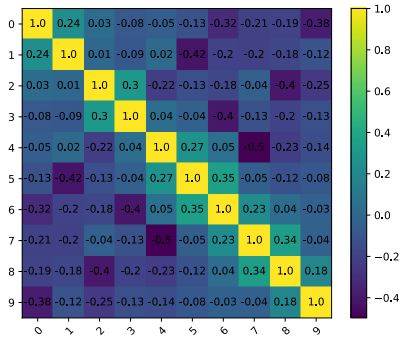


Fig. 6. Pairwise cosine similarity between the learned temporal position embeddings.

AF product provided by NASA FIRMS can be used to acquire labels. But the accuracy of the labeling it produces is limited. Therefore we produce the labels by manually applying thresholds on the original images in combination with a visual inspection. As shown in Fig. 5, the generated manual labeling precisely covers the AF pixels while the VIIRS AF product misses many AF pixels.

V. Result and Ablation Study

A. Setup

We refer to our transformer model and setup as a transformer for AF detection (T4Fire). Each T4Fire model is trained with the same optimizer, Adam, for 50 epochs. All models are also trained with the same learning rate of 0.003 and a weight decay of 0.0003. Because the range of the five channels is different (0–100 for Band I1–I3, 250–400 for Band I4 and I5), the images in the sequence is standardized along its channel by subtracting the variance of the image channel and dividing by the mean of the image channel. Each training sequence has a relatively small memory footprint with input $[x_1, \ldots, x_\tau]^T \in \mathbb{R}^{\tau \times w^2 C}$ and label $y \in \{0, 1\}^\tau$, so the batch size is set relatively large to 512. The Focal Loss [52] is used for training as the vast majority of pixels are nonfire ones and induces an imbalanced label distribution.

1) Architecture of Our Transformer: In Section III we describe and introduce the notation for the transformer model. To review the number of attention heads is denoted by A, the dimension of each feature vector entering and exiting each layer is k, M is the number of the hidden nodes in the MLP defined in (10) and L is the number of self-attention layers applied in the transformer. A grid search of the hyperparameters controlling the architecture was performed to select the best architecture. The number of attention heads is fixed at $A = 3$. The search was conducted over these settings

$$L \in \{4, 8, 12\} \tag{12}$$

$$k \in \{12, 21, 24, 30\} \tag{13}$$

$$M \in \{64, 96, 112, 128, 192, 256\}. \tag{14}$$

Based on the grid search and performance on the validation set, the architecture $L = 4$, $k = 24$, and $M = 112$ were chosen. For Sections V and VI, T4Fire refers to this transformer architecture.

2) Position Embeddings: For the pixel location (x, y), the image patch centered around (x, y) taken from the tth frame in the sequence and flattened is denoted by $x_t \in \mathbb{R}^{w^2 C}$. This vector is then projected by a single-layer network and added to a learned temporal position embedding

$$f_t^{(0)} = \text{Linear}(x_t) + p_t. \tag{15}$$

Fig. 6 displays the pairwise cosine similarity between the learned temporal position embeddings. Close temporal neighbors show a higher level of similarity.

3) Evaluation Metrics: The performance of each model is evaluated using the standard image segmentation metrics of Intersection over Union (IoU) and F1-Score, defined in (16) and (17) for completeness. In the equations, TP, FP, and FN, respectively, represent the number of True Positive, False Positive, and False Negative predictions. IoU measures the

TABLE III

Comparison Between Networks Relying on Spatial and Temporal Information. The Comparisons Involved Are the Network's Size, Speed at Inference Time, and Overall Performance

Model	# Params (M)	Inf. Time (s)	mIoU	F1-Score
Spatial				
PSPNet	2.10	0.0993	0.389	0.537
Swin-U-Net	9.00	0.3930	0.439	0.597
U-Net	14.30	0.0826	0.457	0.609
FPN	13.80	0.0683	0.492	0.647
LinkNet	11.50	0.0948	0.477	0.635
Trans-U-Net	23.50	0.5170	0.476	0.641
Temporal				
GRU-3	0.06	0.5110	0.494	0.647
LSTM-3	0.08	0.5560	0.636	0.756
T4Fire	0.03	0.5800	**0.678**	**0.804**
T4Fire no mask	0.03	0.5450	0.574	0.718
Baseline				
VIIRS AF Prod.	N/A	N/A	0.499	0.663

TABLE IV

Ablation Study on the Effect of Increasing Spatial Information in RNN Models and Proposed Transformer-Based Models

	w = 1		w = 3		w = 5	
Model	**mIoU**	**F1-Score**	**mIoU**	**F1-Score**	**mIoU**	**F1-Score**
GRU-3	0.494	0.647	0.425	0.577	0.421	0.571
LSTM-3	0.636	0.756	0.438	0.581	0.458	0.612
T4Fire	**0.678**	**0.804**	0.557	0.712	0.480	0.643

TABLE V
COMPARISON OF QUANTITATIVE RESULTS OF T4FIRE AND VIIRS AF PRODUCT ON DIFFERENT TEST REGIONS

Fire Name	Coordinates	Start Date	F1-Score		IoU	
			T4Fire	VIIRS AF	T4Fire	VIIRS AF
BC, Canada						
Elephant Hill Fire	(-121.28, 51.00)	2017-07-06	0.828	0.747	0.706	0.596
Eagle Bluff Fire	(-119.50, 49.42)	2019-08-05	0.910	0.789	0.834	0.487
Sparks Lake Fire	(-120.79, 50.92)	2021-06-28	0.815	0.637	0.688	0.467
Lytton Fire	(-121.57, 50.28)	2021-06-29	0.739	0.545	0.587	0.375
AB, Canada						
Chuckegg Creek Fire	(-117.42, 58.38)	2019-05-15	0.794	0.727	0.658	0.571
Sweden						
Ljusdal Complex Fire, Fågelsjö Fire Trängslet Fire, Storbrättan Fire	(14.52, 61.80)	2018-07-14	0.863	0.578	0.759	0.407
New South Wales, Australia						
Currowan Fire	(150.12, -35.50)	2019-11-26	0.799	0.572	0.665	0.400
California, US						
Thomas Fire	(-119.09, 34.42)	2017-12-04	0.803	0.694	0.671	0.532
Tubbs Fire	(-122.63, 38.61)	2017-10-08	0.864	0.669	0.761	0.503
Carr Fire	(-122.62, 40.65)	2018-07-23	0.805	0.558	0.673	0.387
Camp Fire	(-121.43, 39.81)	2018-11-08	0.900	0.714	0.818	0.554
Kincade Fire	(-122.78, 38.79)	2019-10-23	0.831	0.710	0.710	0.550
Creek Fire	(-119.30, 37.20)	2020-09-05	0.779	0.718	0.638	0.560
Blue Ridge Fire	(-117.68, 33.88)	2020-10-26	0.749	0.715	0.599	0.557
Dixie Fire	(-121.42, 39.81)	2021-07-14	0.641	0.695	0.472	0.532
Mosquito Fire	(-120.74, 39.00)	2022-09-03	0.834	0.716	0.678	0.512
New Mexico, US						
Calf Canyon Fire	(-105.50, 35.76)	2022-04-05	0.826	0.703	0.690	0.527
Oregon, U.S.A.						
Double Creek Fire	(-116.73, 45.43)	2022-08-29	0.694	0.531	0.569	0.397

ratio between the area of intersection and the area of union computed between the predicted positives and the ground truth positives. The F1-Score is the harmonic mean of the Precision and Recall. Precision is the ratio between the number of True Positives and all the Positive predictions, i.e., Precision = TP/(TP + FP). Recall is the ratio between the number of True Positives and all the ground truth positives, i.e., Recall = TP/(TP + FN)

$$\text{IoU} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} \tag{16}$$

$$\text{F1-Score} = \frac{2\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{\text{TP}}{\text{TP} + .5\,(\text{FP} + \text{FN})}. \tag{17}$$

As input T4Fire has image patches centered at the same location but from different time-stamps. Its output is a prediction of Fire/Not Fire for each time-stamp of the location under consideration.

B. Quantitative Results

1) T4Fire Comparison to Other Network Architectures: In Table III we report the results of AF detection with approaches using the following models.

1) *Spatial Only Models:* Powerful state-of-the-art ConvNet and transformer architectures are applied independently to each individual input.
2) *Temporal Models:* RNN-based approaches and our proposed T4Fire approach. All these networks use the same simple temporal sequences as input.
3) *Baseline:* The VIIRS AF Product.

We chose the Temporal Models and the Spatial Only Models to investigate and quantify the benefit of relying mainly on temporal information and local spatial information over-relying on more global spatial information but with no temporal information. For the latter, we trained very big and high-capacity networks that have shown high performance for semantic segmentation for a range of application domains such as medical imagery and imagery from self-driving cars. We also wanted to investigate the effectiveness of T4Fire at exploiting temporal information over RNN-based networks. All the performance numbers quoted in the table are the average of three training runs.

All the datasets used by the Spatial Models are standardized using the same methods as the T4Fire model. The spatial models have ResNet-18 as their backbone and use pretrained weights from the ImageNet dataset. The training minimizes the Dice Loss [53] function and is optimized by Adam with a learning rate of 0.0001 and batch size 32. The inference time shown in Table III is the inference time of the pretrained model on one NVIDIA Tesla T4 GPU from the Google Colab Platform.

The RNN-based Temporal Models, shown in Table III, have three layers and each hidden layer of the LSTM and GRU models has a size of 64. Parameters are determined by running a parameter grid-search in parameter space

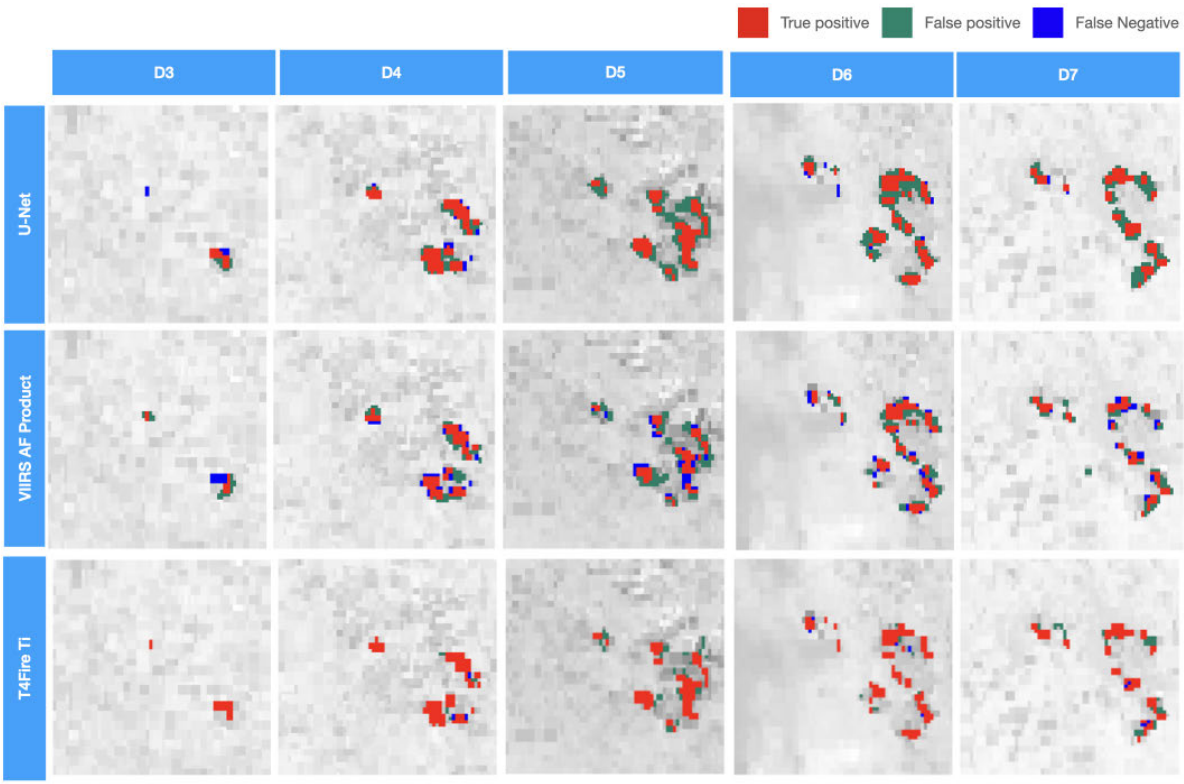


Fig. 7. Qualitative results of the Ljusdal Complex Fire, zoomed-in to the center of the fire. The AF pixel detection results are shown for three approaches: U-Net with ResNet-18 backbone, VIIRS AF Product, and T4Fire. The predicted detections for each time-step are overlaid on the Band I4 image. True positive detections are shown in red, false positives in green and false negatives in navy. The average F1-score achieved for this fire is 0.863 for T4Fire, 0.585 for U-Net, and 0.759 for VIIRS AF Product. Images of D_1- and D_2 and D_8–D_{10} are not shown because there are no AFs in those days.

$n_{\text{hidden}} \in \{32, 48, 64, 96, 128, 256\}$ and $n_{\text{layers}} \in \{2, 3, 4, 5\}$. Both Temporal Models are trained using Focal Loss and an Adam optimizer with a learning rate of 0.003 and batch size 512.

The results clearly show that T4Fire, with $w = 1$, has the best segmentation result of all the networks investigated. For the T4Fire model without masked self-attention, both IoU and F1-Score are slightly lower than the masked T4Fire. This indicates that masked self-attention helps the transformer to better capture the temporal dependencies in the input sequence. Although, the masked self-attention slightly increases the inference time as shown in Table III because it adds an additional computation step. All of the networks from the spatial-only model grouping, except for PSPNet which performs significantly worse, show similar performance and are outperformed by the temporal models for AF detection. In fact, these models do not even perform better than the baseline model. As these spatial models are very large and have a high capacity it may be that we did not have sufficient labeled training data and computational resources to adequately train them. But it is more likely that spatial correlations and dependencies are much less in VIIRS imagery than in other images and thus networks relying on these correlations are not as effective for AF detection. Due to the spatial resolution of VIIRS images each of its pixel roughly represents a 375 × 375 m area on the ground. The low spatial resolution makes the correlation between two neighboring pixels low because the land cover, soil moisture, terrain, or even precipitation and temperature of such a large area can drastically change. Thus, ConvNet-based spatial models which use spatial correlations are not as useful in the classification. From Table III, ConvNet-based methods show significantly faster inference speed than all the temporal models and transformer -spatial models like Swin-U-Net and Trans-U-Net. However, since all models have an inference time of less than 1 s/image, the impact of inference time on the application is trivial. In Section V-C the qualitative results of study areas are displayed and they indicate that the low metrics at times are caused by high false positive rates.

For the Temporal model grouping, both GRU and LSTM models outperform the Spatial-only-based models but their performance is worse than T4Fire. Of interest is also that all the temporal models have orders of magnitude fewer parameters than the Spatial networks with T4Fire having the fewest. Compared to spatial models, temporal models use temporal changes over the same location. The high temporal resolution but the low spatial resolution of VIIRS makes temporal models outperform spatial models. In summary temporal information appears to be much more powerful in detecting AF pixels for VIIRS imagery than spatial information alone and a vanilla transformer architecture can outperform vanilla LSTM and GRU architectures.

Finally, the VIIRS AF Product, the baseline method used in the real world today, gives worse coverage of AF according to the quantitative results. In Section V-C, the qualitative results indicate the AF product commonly underestimates and overestimates the AF region compared to T4Fire.

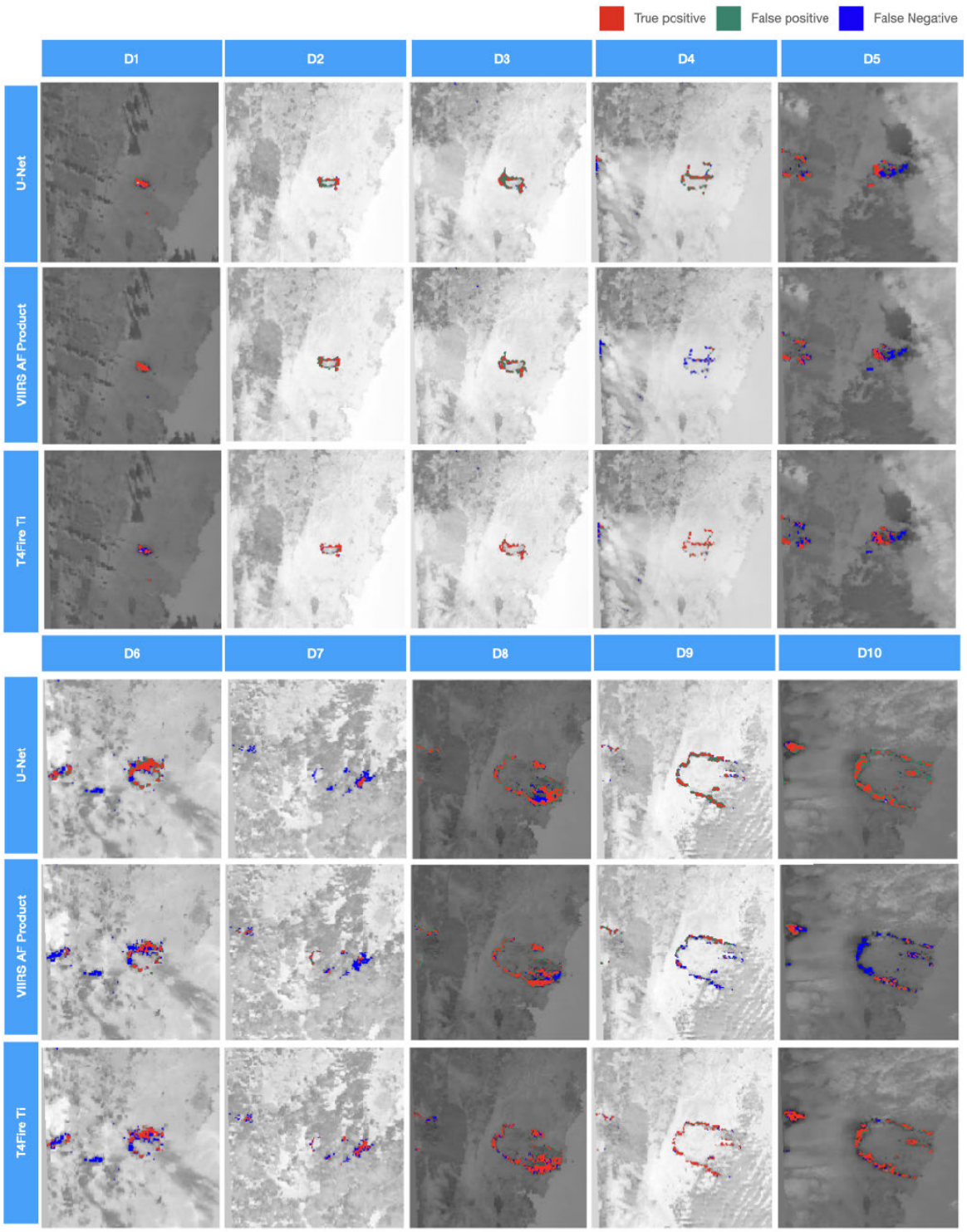


Fig. 8. Qualitative results for the Currowan Fire. The AF pixel detection results are shown for three approaches: U-Net with ResNet-18 backbone, VIIRS AF Product, and T4Fire. The predicted detections for each time-step are overlaid on the Band I4 image. True positive detections are shown in red, false positives in green and false negatives in navy. The average F1-score achieved for this fire is 0.799 for T4Fire, 0.643 for U-Net, and 0.572 for VIIRS AF Product.

2) T4Fire's Performance on Individual Wildfires: Table V reports the quantitative results for different regions and individual wildfires. The results indicate the T4Fire network provides stable results across different study regions. The T4Fire network, which relies on temporal information, seems to be able to capture the pattern of fluctuation of thermal radiation present for wildfires across regions. The VIIRS AF Product also provides quite stable results across the different test regions, though the F1-Score and IoU are significantly lower compared to those obtained by T4Fire.

3) Ablation Study: In the prior quantitative results, the patch extracted from each pixel had size $w = 1$. In this set of experiments, we investigate if increasing the local spatial information in the temporal networks helps. In particular,

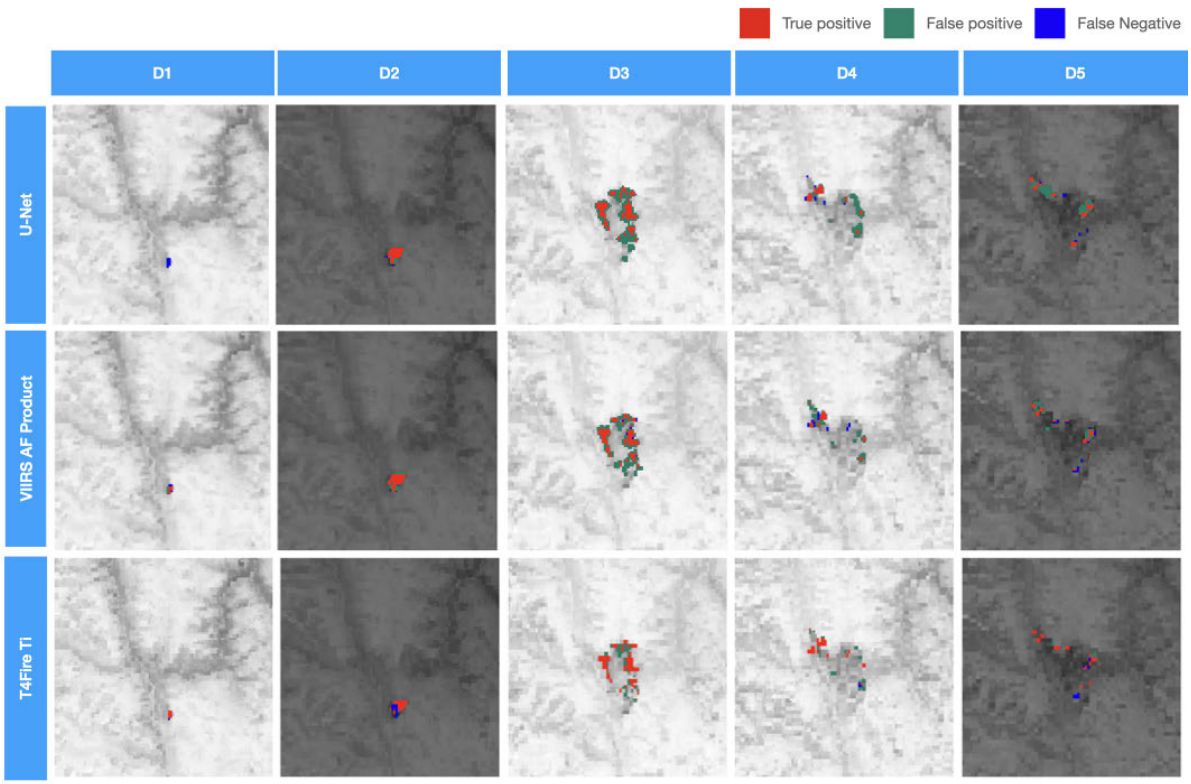


Fig. 9. Qualitative results for the Lytton Fire, zoomed-in to the center of the fire. The AF pixel detection results are shown for three approaches: U-Net with ResNet-18 backbone, VIIRS AF Product, and T4Fire. The predicted detections for each time-step are overlaid on the Band I4 image. True positive detections are shown in red, false positives in green and false negatives in navy. The average F1-score achieved for this fire is 0.739 for T4Fire, 0.499 for U-Net, and 0.587 for VIIRS AF Product. D_6–D_{10} are not shown because there was no significant fire.

we use the spectral values of the patch's single pixel but also its neighborhood pixels that is we set $w > 1$. Three image patch sizes 1×1, 3×3, and 5×5 are tested for T4Fire and the RNN variants as shown in Table IV. Remember when $w = 1$ the image patch size equals 1 and no neighboring pixels are used. It can be observed that increasing the window size causes a significant performance drop for T4Fire, GRU-3, and LSTM-3. This further indicates that using temporal information is more important than spatial information.

4) Summary of the Quantitative Experiments: The quantitative results lead to conclusions.

1) The T4Fire model performs best of the spatial and temporal networks investigated. Also, the proposed method shows better quantitative results than the baseline VIIRS AF Product.
2) The temporal information of the VIIRS images contributes more than the spatial information in detecting AF pixels. The variation of the spectral values helps to identify the AF pixels more reliably and accurately than the spatial correlations in the neighborhood spectral values.

It would be interesting to explore in further work if there is an architecture and/or patch embedding that would allow a better exploitation of the spatial and temporal information for active wildfire detection.

C. Qualitative Results

In this section, the time-series of images from two wildfires are tested to generate the AF segmentation. To better understand the segmentation result, the output is overlaid on the Band I4 thermal images in which the rough shape of the AF can be visually observed. The qualitative results are shown for our novel T4Fire network as well as the relevant and common baselines of a simple spatial-only U-Net and the VIIRS AF Product.

1) Study Area 1: The 2018 Forest Fires in Sweden: In the summer of 2018, over 50 forest fires burned across Sweden, from north of the Arctic Circle to the southernmost county of Scania. The test dataset includes four of the largest 2018 wildfires, i.e., Ljusdal Complex Fire, Fgelsj Fire, Trngslet Fire, and Storbrttan Fire that burned a total of 17 330 hectares. The relatively small size of the wildfires and their fast spread speed make the detection of AFs difficult using coarse-resolution satellite images. For the qualitative results, only the Ljusdal Complex Fire is visualized in Fig. 7.

The proposed T4Fire network could detect the majority of AF points in the images of Ljusdal Complex fire, except in D_5 and D_7 where there is a slight overestimation in the center part of the image. Comparably, the VIIRS AF Product both overestimates and underestimates some parts of the wildfire at D_5 and D_7. Both the U-Net and VIIRS AF Product produce several false positive detections in D_6 and D_7. And these false positives lead to their F1-Scores being lower than T4Fire's for this region.

2) Study Area 2: Currowan Fire, New South Wales, Australia: Broke out on November 26, 2019, this wildfire burned a total of 4 99 621 hectares of vegetation during

74 days. Due to its large size, only one of the regions affected by the wildfire is used as the study area.

As shown in Fig. 8, VIIRS AF Product does not detect the AF pixels at D_4, D_7, and D_{10}. Especially at D_{10}, a large portion of the wildfire is missed in the image. Consequently, VIIRS AF Product only has an F1-Score of 0.572. The predictions for both T4Fire and U-Net cover the AF pixels very well. The U-Net though does over-estimate at the boundaries of the wildfire and thus has a slightly lower F1-Score than T4Fire.

3) Study Area 3: Lytton Fire, British Columbia, Canada: Started on June 30, 2021, the Lytton fire is one of the most devastating fires in British Columbia, burnt a total of 83 671 hectares including the town of Lytton.

As shown in Fig. 9, the proposed T4Fire offered good predictions of AFs in the majority of the timestamps. The U-Net overestimates AFs on multiple timestamps. Overall, T4Fire achieved a much higher average F1-Score (0.739) and IoU score (0.587) than the F1-Score (0.545) and the IoU score (0.375) for the VIIRS AF Product. Visual inspection of the qualitative results also confirms this. Compared with T4Fire and VIIRS AF Product, U-Net produces more false positives, especially in D_3–D_5. It also confirms temporal information represented by the variation of band values over time is more useful in detecting AF pixels than spatial information.

VI. Conclusion

In this research, we proposed a transformer network-based AF detection model using VIIRS image time-series. Based on the quantitative and qualitative results on the test regions, the proposed network significantly improves the AF detection accuracy compared to the multicriteria threshold-based AF product from VIIRS. We also compared the performance of temporal to purely spatial deep learning models and the results indicate that temporal deep learning models have superior performance for AF detection. This shows the fluctuation of the spectral information in the time domain is a better indicator of the AF pixels than the spatial information alone. Additionally, we compared our proposed transformer network with other RNN models. Even though all temporal deep learning models perform very well in detecting AF pixels, the transformer network shows much better quantitative results. In an ablation study, we include more neighboring pixels as the input of the transformer network to use both temporal and spatial information. The results further confirm that only using the temporal information achieves the best performance.

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